Anushika Malviya

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Summary

Data Analyst and Machine Learning student (Integrated M.Tech CSE, VIT Bhopal, CGPA 7.68/10) who built NLP, recommendation, and RAG systems. Developed 3 machine learning projects and 2 MERN web applications using Python, SQL, scikit-learn, and React.

Education

VIT Bhopal University — Bhopal, India 2022 – 2027
Integrated M.Tech. in Computer Science and Engineering (Computational and Data Science) *CGPA: 7.68/10*

Technical Skills

Languages: Python, SQL, C++, JavaScript
Machine Learning & AI: Machine Learning, NLP, scikit-learn, TF-IDF, Cosine Similarity, RoBERTa, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Evaluation
Data Analysis & Visualization: Data Analysis, Data Visualization, Tableau, Microsoft Excel, Plotly, Trend & Feature Analysis
Web & Backend: MongoDB, Express.js, React.js, Node.js, Flask, FastAPI, Streamlit, REST APIs, JWT Authentication
Tools & Practices: Git, GitHub Actions (CI), Docker, Agile, Code Review, Cross-Functional Collaboration, Data Structures & Algorithms

Experience

MERN Full Stack with AI Internship Program Nov 2025 – Dec 2025
Intern *Remote*

- Integrated 3 AI features (recommendations, workflow automation, data insights) into 2 web applications during a 6-week MERN internship.
- Architected and deployed 2 web applications with 10+ REST API endpoints, secure JWT authentication, and managed MongoDB databases.
- Collaborated with engineering and design teams on feature development using code review, Git version control, and Agile sprints across 2+ cycles.

Projects

DocuMind | Python, FastAPI, Streamlit, Docker, GitHub Actions | GitHub

- Built a RAG pipeline with local embeddings and 2-stage retrieval (bi-encoder search + cross-encoder reranking), supporting 2 LLM providers and a retrieval-only mode.
- Cut hallucination risk by grounding answers strictly in retrieved passages and enforcing source citations.
- Benchmarked retrieval with Recall@k and MRR on a labeled question set, and scored answer faithfulness and relevance with an LLM judge.
- Shipped 2 interfaces, a FastAPI REST service and a Streamlit chat UI, containerized with Docker and tested via GitHub Actions CI on every push.

Sentiment Analysis Dashboard | Python, RoBERTa, TF-IDF, Streamlit, Plotly | GitHub

- Classified 600 sample customer reviews into 3 sentiment classes (positive, neutral, negative) with RoBERTa, with a lighter DistilBERT option for faster inference.
- Surfaced the drivers of negative feedback via TF-IDF keyword ranking, with configurable-threshold alerts for negative-sentiment spikes.
- Architected a 3-layer codebase (preprocessing, model, analytics) behind a Streamlit + Plotly dashboard with 4 insight views, CSV upload, and filtered-data export.

AI-Powered Course Recommendation System | Python, scikit-learn, React, Node.js | GitHub

- Built a content-based recommender (TF-IDF, cosine similarity) over a 25+ course catalog with a 5-factor hybrid score ($S = 0.40T + 0.30K + 0.15D + 0.10P + 0.05F$) blending text similarity, skill gaps, difficulty, preferences, and feedback.
- Raised Precision@3 from 0.20 to 0.53 (+166.7%) over a popularity baseline; NDCG@5 of 1.0, MAP of 0.78.
- Served real-time predictions to a React frontend via a Flask microservice and 23 JWT-secured REST endpoints, with skill-gap analysis across 6 career roles.

Certifications & Achievements

- Machine Learning — Coursera
- Machine Learning — NPTEL
- AI Skills Passport — Microsoft & EY
- Solved 100+ DSA and SQL problems on LeetCode and Codeforces across data structures, graph algorithms, and joins.